

Toxicity in Large Language Models: An Ethical Reflection

January 2025
Word count: 759

1 Project Aim

The aim of this project is to identify toxicity in the output of Large Language Models (LLMs) that are given toxic inputs, or prompts. The degree of toxicity (or threshold) is measured using the *Perspective API* [3]. The PyTorch library *Captum* is used to analyse and understand the given outputs of the LLMs [2]. Since the project is an analysis, this is what is referred to when mentioning the term "product".

2 Ethical Problems in the Project

Some ethical problems that were known that could arise in the project are:

- Biased training datasets, which can result in biased models.
- Lack of transparency in the model's decisions and outputs.
- Dependency on a few companies that control toxicity detection tools.

Another ethical problem that could arise, which wasn't discussed initially but is highly relevant for the project, is the usage of LLMs with malicious intent. AI-powered social media bots could (semi-)autonomously harm society. For example, they can create and spread fake news or escalate online discussions, leading to unwanted polarisation [1]. The latter functionality is especially relevant because toxic online comments can provoke others to respond toxically as well, potentially spiralling into increased polarisation.

As the product aims to provide insight into why and how toxicity is generated by LLMs, it could be used by systems like ChatGPT-4.0 to detect toxic inputs and prevent the model from generating toxic outputs. While the product has the potential to decrease toxic generations if investments are made to deepen and expand the research, it could also be used to circumvent the implementations made by tech companies. Therefore, an ongoing conflict might arise between

developing better preventative techniques and finding ways to circumvent these same techniques. This tension is especially concerning when considering the potential for polarising chatbots using the product.

During the project it was also found that the Perspective API contains biases when scoring the toxicity. A neutral statement concerning an ethnic minority for instance led in many cases to an increased toxicity score.

3 Ethical Research

A stakeholder analysis with a utilitarian approach is used as the research method, as the potential effects of the product on society as a whole are weighted the heaviest. Direct stakeholders to the product are developers and users of LLMs, both privately and commercially. Indirectly, however, whole societies could also be affected by the toxicity of LLMs, as described in the previous section.

4 Suggestions for Potential Solutions and Future Directions

Developers of LLMs could integrate the findings of this project into their systems, leading to improvements in toxicity detection and prevention mechanisms. This would create not only a safer user experience, but more importantly a transparent one. It is suggested that on a small scale users should be able to work with potentially toxic outputs (to a certain threshold), as censoring them completely is not recommended. For this to be considered ethical, it is pivotal that if an LLM outputs something that could be considered toxic that it provides the user with a trigger warning. If this is implemented users are made aware that some things they might not consider to be toxic could be for instance prejudiced or offensive. Therefore, it is not recommended that they completely stop generating outputs, since the user might not have realised that the prompt could lead to a toxic output. On the other hand, on a larger scale the LLMs should stop generating as people with malicious intentions could use the outputs for societal, harmful activities. Thus the number of toxic outputs an LLM will be able to generate is an ethically motivated trade-off between providing users with what they ask for and maybe educating them versus preventing malicious activities on a large scale. If fully implemented, this should result in a Value-Sensitive Designed product.

5 Evaluation of Ethical Process

The suggestion that LLMs should provide a warning to potentially toxic outputs is based on the fact that future LLMs can identify toxic outputs. Besides, they should provide the user with a nuanced description of sensitive topics. At the moment humans are way more capable of doing this task than LLMs, so this

application is heavily based on the assumption that LLMs will improve to the point that they could analyse prompts ethically superior (what ethically superior means might be interesting for future research). As the Perspective API contains biases in the way it scores prompts and outputs on toxicity, further research to diminish these biases is essential for future LLMs to be value sensitive. The choice of a utilitarian approach inherently neglects other considerations. One could argue, for instance, that everyone should be able to say anything they want on social media. Not considering facts or the toxicity of their statements.

References

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